

Review

THE ROLE OF PHARMACOGENETICS / PHARMACOGENOMICS IN CLINICAL PRACTICE

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Information about Drug metabolizing enzymes (DME) capacity might be helpful in clinical practice to optimize pharmacotherapy and may reduce risks of unwanted drug reactions e.g. in psychiatry or cardiology (Arranz *et al.*, 2000; Chen *et al.*, 1996; Eichelbaum & Gross, 1992). Adverse drug reactions included mortality and morbidity are important from a health care point of view as well from a pharmaco-economics consideration. Adverse drug reactions (ADR) usually associated with significantly prolonged stay in the hospitals, increased economic but and an almost 2-fold increased risk of death (Classen *et al.*, 1997).

A US data shows serious adverse drug reaction (ADRS) occur amongst 6.7% of all hospitalized patients and almost 0.32% of them develop fatal adverse reactions, causing more than 100,000 deaths annually indicating a 4th to 6th leading cause of death and causing 100 billion USD (Lazarou *et al.*, 1998).

Though this figure has been criticized for over estimation still other studies find similar data in other European-countries. Data from Sweden indicates a 12% hospitalization due to ADRs. While estimating 4% as preventable ADRs (Mjorndal *et al.*, 2002). The clinical consequences of impaired metabolism have been studied extensively during recent years (Edeki, 1996; Spina & Caputi, 1994; Zuhisdorf, 1998; Wang *et al.*, 1993; Seivastawa, 2003 and Meyer & Zanger, 1997).

For certain psychotropic drugs it is documented that 20-30% of patients do not respond to therapy (Meyer & Zanger, 1997). Therapeutic drug monitoring (TDM) may decrease the proportion of non-responders to 10-20% indicating a relationship between blood levels and clinical effect (Weber, 1999).

ADRs that are considered as non-preventable include severe allergic reactions to antibiotics in patients without a previous history and acute renal failure, associated with drugs, in critically ill patients. Known risk factors for ADRs include age, polypharmacy, liver and renal diseases as well as gender. Identifying factors, as predisposing to ADRs, is essential for risk management. For a monogenic genetic polymorphism, predicting genotyping or phenotyping for known genetic polymorphism was estimated to avoid the occurrence of 10 to 30% of ADRs (Bush, 1998 and Ingelman, 2001).

It is expected from pharmacogenetic/ pharmacogenomics disciplines that consider the patient's genetic information of drug transporters, drug metabolizing enzymes and drug receptors, to allow for a targeted and individualized efficient drug therapy leading to optimal drug choice and dosage regimen of the drugs, as well as, reducing cost of treatment (Meyer *et al.*, 1996).

Drug Metabolism:

Seivastawa (2003) recently reviewed in

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detailed drug metabolism and its impact on drug response and approach towards individualized medicine. Author describes that Drug metabolism refers to the biochemical transformation of a compound into another more polar chemical form. Absorption, distribution, metabolism and excretion comprise an integral part in understanding the safety and efficacy of a potential new drug. Detailed in depth knowledge of the Pharmacokinetics and Drug Metabolism of a new drug entity is considered a prerequisite to know the appropriate route of administration, correct dose etc. Sometimes there is (are) different/unwanted effect(s) of certain drugs in different populations. This is particularly true for the drug having narrow therapeutic index. Often these different effects are detrimental to an individual, thus termed as adverse drug reactions. After the raw draft of human genome has revolved, it has become increasingly clear that change(s) in the drug response between individuals, is due to the occurrence of genetic polymorphisms in the Phase I and II drug metabolizing enzymes, due to which distinct subgroups in the population differ in their ability to perform certain drug bio-transformation reactions. The study about the occurrence of genetic polymorphisms in drug metabolizing enzymes is termed as Pharmacogenetics/Pharmacogenomics. Pharmacogenetic characterization of particular drug can be either phenotypically or genotypically conducted in population groups. Such a study is very important to check the post-marketed drug withdrawal, if a particular percentage of population suffers from adverse drug reactions, and thus a lot of revenue be saved. Such studies also help to find out Right Medicine for Right Individual or Individualized Medicine (Seivastawa, 2003).

Factors Affecting Drug Metabolism:

Many factors can influence the drug metabolism enzyme activities and thus metabolism of drug and environmental chemical through these enzymes.

Therefore, pharmacokinetic and pharmacodynamics of drugs are affected by internal

factors as well as external factors internal factors include physiological and pathological (factors as sex, age, hormonal status, genetics, disease and life style). While the external factors include diet and nutritional state, tobacco smoking, environmental chemicals, occupation, drugs, socio-economic states, cultures, etc.

Some drugs and chemicals can influence metabolism of other drugs and chemicals. Similarly certain individuals and populations (ethnic groups) can have different forms and level of these enzymes and therefore result differently to the same drugs (Meyer & Zanger, 1997; Weber, 1999 and Pande *et al.*, 2003).

Molecular Mechanisms of Genetic Polymorphisms of Drug metabolism:

A comprehensive review has been presented by Meyer and Zanger (1997). Meyer explained that mainly the genetic variation of drug metabolism causes the inter-individual variation in drug effects. Based on the genetic polymorphism of the drug metabolizing enzymes, distinct population subgroups are described, differing in their ability for certain drug bio-transformation reactions. Polymorphisms generated by mutations in the gene responsible for these enzymes, causes decreased, increased or absent enzyme expression or activity by multiple molecular mechanisms.

The variant allele usually occurs at high frequencies in populations. Molecular mechanisms of these polymorphisms are of major interest. These polymorphism include two major groups:

Acetylation Polymorphisms; concerning the metabolisms of a certain arylamine and hydrazine drugs and carcinogens, by the cytosolic N-acetyltransferase (NAT2). Polymorphism is exhibited as rapid, intermediate and slow metabolizers, being homozygous dominant, heterozygous dominant and homozygous recessive, respectively.

Polymorphism of P450 enzymes, concerning the polymorphisms of drug oxidation, affecting metabolism of more than 40 drugs. Poor metabolizer exhibit loss of functional allele of cytochrome P450 CYP2D6 gene, ultra rapid is caused by duplication or amplification of active CYP2D6 gene; while intermediate metabolizer are often heterozygous or carry alleles with mutations with moderately decreased enzyme activity (Meyer & Zanger, 1997).

Inter-individual differences, of clinical significance, in response to drugs and environmental and dietary chemicals, are caused by genetic and environmental factors. Most often this "pharmacological individuality" is due to variability of drug metabolisms in the liver.

Major contributor of this Variability are, the polymorphism of the genes coding for drug metabolizing enzymes, and transcriptional activation of these genes by drugs and environmental factors.

Recently the emphasis is on the understanding of the molecular mechanisms of variability in human drug response and diseases. The ideal goal is to develop, simple, non-invasive methods by which individual at risk develop adverse drug reactions or by which drug/or chemical induced diseases can be recognized.

Molecular Mechanism of three most common genetic polymorphism of drug metabolizing enzymes are (Meyer & Zanger, 1997):

- 1) CYP2D6 polymorphism: 5-10% of Caucasians poor metabolizes of Debrisoquine & 40 other drugs.
- 2) CYP2C19 polymorphism: 2-5% of Caucasians but 20% of Orientals; poor metabolizes of and 20 other drugs.
- 3) N-acetyltransferases: 40-70% of Caucasians are slow acetylators of numerous drugs and chemicals.

These polymorphism are transmitted as

autosomal recessive traits and cause impaired inactivation of numerous clinically important drugs or chemical carcinogens in "poor metabolizer" individuals.

For each of these polymorphism, the target enzymes have been purified from human liver, the normal and the mutant genes cloned, sequenced and expressed and the mutations causing these variations identified.

Identification of over 95% of all mutations of the CYP2D6 gene and NAT2 genes has been done. New DNA based chip technology is a promising technique that can evaluate multiple mutations in a single DNA sample (Meyer & Zanger, 1997).

In addition to drug metabolism enzymes, Pharmacogenetics is recently been extended to drug transporters, receptors, and targets that can modulate drug response [about 0.1% of human genome sequence is polymorphic]. The goal for Pharmacogenetics is to determine clinical consequences of 2-3 million SNPs. These could be divided basically into two as:

- (i) Frequent SNPs (allele frequency > 10%) which have a low impact on drug response (Odds ratio <2), even combined with other SNPs in haplotype combinations.
- (ii) Rare SNPs which alter the expression or activity of a target protein, can sometimes have a real clinical relevance (Odds ratio >5) usually to predict drug side effects (e.g. TPMT & CYP2C9), less than 1 % of population is concerned by these rare SNPs (Meyer & Zanger, 1997).

Populations and Genetic Polymorphism:

Dependent on the ethnicity or ethnic characteristic of a population, many polymorphic genes of pharmacogenetic interest, may support the evaluation of individual differences in drug effectiveness (hypersensitivity or resistance) and drug toxicity.

Weber in 1999 described that ethnic specificity is an integral part of Pharmacogenetics because it can suggest a starting point for further study of the trials, tailoring

drug therapy to the individual patient on one hand, and rational development and clinical trials of new drug on the other. He expressed many such illustrations of ethnic specificity of clinical importance (Wang *et al.*, 1993).

Pande in 2003 recently reviewed the polymorphism of acetylation metabolism and Individual susceptibility to diseases due to several allelic variations for both NAT1 and NAT2. The ethnic distribution supports an interpretation that the acetylation polymorphism existed before Paleolithic splitting of human populations from Africa (Lin *et al.*, 19947).

Wilson *et al* in 2001 explained the geographic patterns of genetic variations, including variation at Drug Metabolizing Enzyme (DME) Loci and drug targets, indicate that geographic structure of inter-individual variation in drug response may occur frequently. They found commonly used ethnic labels are insufficient and inaccurate.

EDP approach was adopted by Nebert and Dieter in 2000, who explained that in the identification of responsible gene, there is a gradient in phenotypes (efficacy or toxicity), it is therefore important to select a patient who shows definite phenotype possible. This is the method of Extreme Discordant Phenotype (EDP), which he highly recommended.

Michel in 2003, recently discussed the controversy of place of origin or ethnic affiliation proxy, which weekly resemble the distribution of neutral genetic variation, affecting drug response. Using Alu insertion on samples from all 3 continents, of 565 individuals, they showed that these proxies correspond crudely, if at all, to population structure, suggesting a more flexible frame work for using such proxies.

Population frequencies of many polymorphic genes of pharmacogenetic interest depend on race or ethnic specificity. Association of these genes with person-to-person differences in drug effectiveness

(hypersensitivity or resistance) and drug toxicity may also depend on the racial or ethnic characteristics of a population. Information about ethnic specificity is an integral part of Pharmacogenetics because it can suggest a starting point for further study of these traits, tailoring drug therapy to the individual patient, and rational development and clinical trials of new drugs. Ethnic specificities of several medically important metabolic traits serve to illustrate these ideas. Among the traits considered is primaquine sensitivity, a sex-linked trait attributed to glucose-6-phosphate dehydrogenase deficiency that mainly affects males among African, Mediterranean, and Oriental people. Additional examples include the remarkable sensitivity of the Japanese to alcohol (ethanol) compared with whites; the ethnic specificity of the cytochrome enzyme CYP2D6* (debrisoquine/spartein) polymorphism that results in poor, extensive, and ultra rapid metabolizers of at least 30 drugs; the CYP2C19* (mephenytoin) polymorphism that accounts for variable metabolism of proguanil, omeprazole, and certain barbiturates; and the polymorphic (NAT2*) acetylation of hydrazine and aromatic amine drugs, such as isoniazid, hydralzaine, and sulfasalazine (Breimer and Danhof, 1997).

Genetic polymorphism and Environmental Health:

The role of variations in the human genome (genetic polymorphism in altering the effects of exposure to environmental Health hazards (gene-environment interaction), that render some individual or groups in the population more or less likely to develop diseases after exposure. This incorporation of genetics and environmental health is a rapidly growing research area, primarily to evaluate the ability to detect different level of risks within the population and to understand the etiologic mechanisms involved (Timbrell, 1985).

Drug Metabolizing Enzymes (DME):

Humans are continually exposed to foreign chemicals. Drug metabolizing

enzymes are responsible for the biotransformation and excretion of foreign chemicals, such as drugs, from the body, after their desired effect has been exerted in human; by converting them into more polar and water soluble products that has reduced affinity to bind to particular target proteins like enzyme, ion channel or receptors) and that could more easily be excreted via urine or bile (Rowland and Tozer, 1994).

The intensity of drug effect (pharmacodynamics) depends on the concentration of the drug in direct environment (biophase) of the target and the affinity of the drug to the target. As all organs and tissues are supplied with blood, therefore drug concentration in blood-time profile (pharmacokinetic) determines its effectiveness (pharmacodynamics). The basic process of absorption, distribution and elimination are responsible for the transport and distribution of drug molecules and thus influences there over all behaviors in the body (Rang *et al.*, 1995 and Nebert *et al.*, 1999).

After absorption drug reaches to the systemic circulation. The extent of absorption (bioavailability) determines the fraction of the dose accessible to the systemic circulation. Bioavailability may also be influenced by metabolism in the gut and in liver (first-pass) before reaching systemic circulation. From the systemic circulation the drug is distributed in various body compartments, the four main compartments are plasma water (upto 5% of body weight), interstitial fluid (upto 16%), intracellular water (upto 35%) and fat (upto 20%) (26). It follows simultaneous elimination from body, main processes being, excretion and biotransformation. Main route of excretion are urine (renal excretion) bile feces (hepatobiliary excretion).

Enzymatic biotransformation (metabolism) occurring mainly in liver and also in other tissues, reduces the lipid solubility of a drug, making it polar, to be more water soluble for easy excretion. Biotransformation can alter the biological effects of drugs and other

xenobiotics.

Phase I, II and III Enzyme Systems:

Biotransformation reactions are generally divided into two steps, as phase-I and phase-II (Rowland & Tozer, 1994; Rang *et al.*, 1995; Nebert *et al.*, 1999 and Benet, 1997). In phase-I, (a functionalization reaction) chemical modification of the parent drug takes place, usually involving cleavage of alkyl group or insertion of polar group into the non polar parent compound by oxidation, reduction, hydroxylation, hydrolysis, etc. In phase-II, basically a conjugation reaction, the polar groups of the intermediate metabolites, arising from phase I or parent drug, conjugated to endogenous highly water soluble groups, such as glutathione, glucuronic acid, amino acid or acetylated or sulphate etc., rendering the molecule for easy excretion. Phase I reaction responsible enzymes include the oxidases – (mainly Cytochrome (CYP) P450, flavin monooxygenases, xanthine oxidase), reductases, deaminases and hydrolases, such as epoxide hydrolase. CYPs are the main and the most important DME among phase I enzymes. Phase II responsible enzymes include glutathione-S-transferase (GST), Sulfatransferase, N-acetyltransferases (NATs), UDP-glucuronosyltransferase (UGTs) etc. (Fig. 1).

Phase III enzyme System:

Antiporter activity (p-glycoprotein or multi-drug resistance) has been recently defined as the Phase III detoxification system (Benet, 1997 and Liska, 1998).

Antiporter activity is an important factor in the first pass metabolism of pharmaceuticals and other xenobiotics (28).

In the intestine antiportal activity appears to be coregulated with intestinal Phase I CYP 3A4 enzyme. It is this observation that suggests antiporter may support and promote detoxification. By pumping non-metabolized xenobiotics out of the cell and back into the intestinal lumen may allow more possibility to Phase I enzyme activity to metabolize the xenobiotic, before it move into circulation.

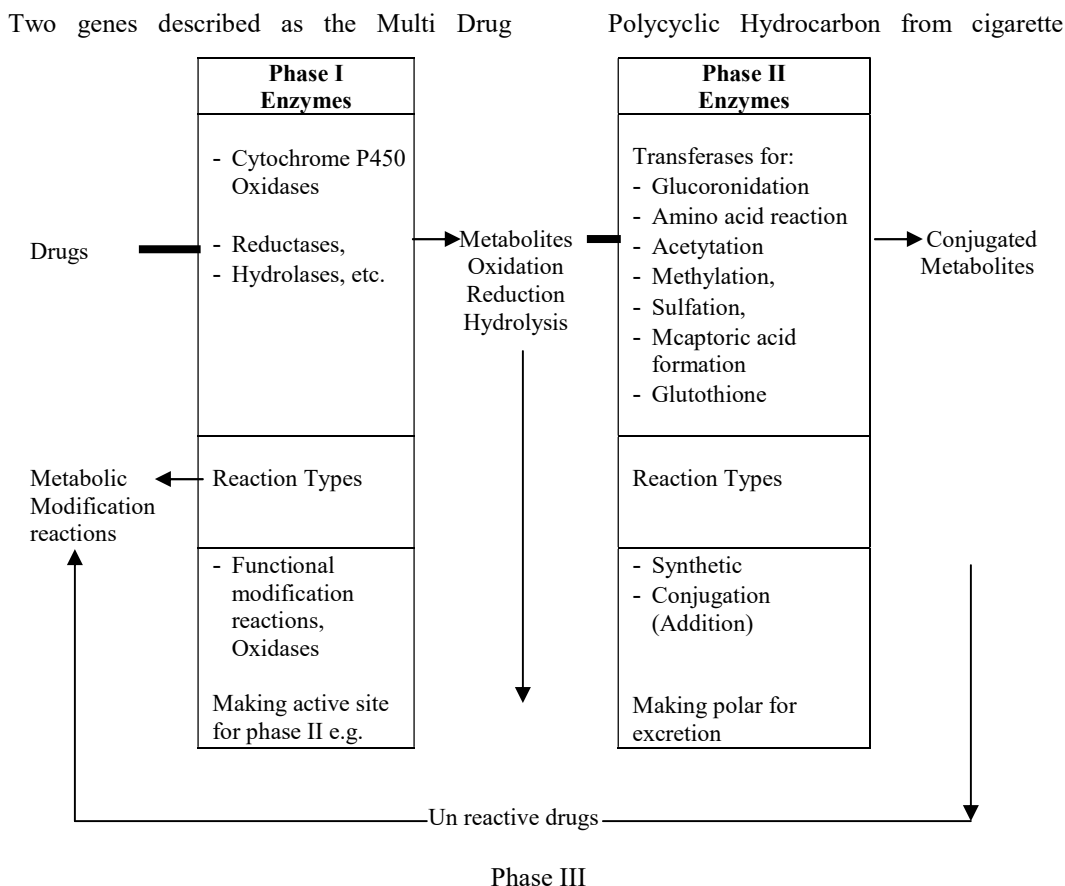


Fig. 1: Phase-1, Phase-II and Phase-III Drug Metabolism.

Resistance gene 1 (MDR1) and Multi Drug Resistance Gene 2 (MRR2); encoding for antiporter activity (Watcher *et al.*, 1995).

Body has developed several mechanisms to regulate detoxification activity. Specific detoxification pathways may be induced or inhibited depending on the presence of various dietary or xenobiotic compounds, the age and sex of the individual, genetics, and lifestyle habits, like smoking. Diseases can also influence activity of the enzymes, induced upgraded or inhibited or reduced (Weaver & Hedrick, 1997; Keeton, 1980; Stryer, 1981; Hein *et al.*, 2000; Kalow *et al.*, 1986 and La Du 1992).

smoke and aryl amines from charbroiled meats, result in dramatic induction of CYP1A1 and CYP1A2 enzymes, leading to a substantial increase in Phase I activity, with little or no induction of Phase II enzymes (Weaver & Hedrick, 1997; Keeton, 1980; Stryer, 1981; Hein *et al.*, 2000; Kalow *et al.*, 1986 and La Du 1992).

Induction of Phase I activities without induction of Phase II activities may lead to an uncoupling I and Phase II balance of activity, and therefore a higher level of reactive intermediates which can cause damage to DNA, RNA and proteins (Hein *et al.*, 2000; Kalow *et al.*, 1986 and Price Evans *et al.*, 1960).

Table 1
Pharmacogenetics of Phase-I and Phase-II Drug metabolism

A. Pharmacogenetics of Phase-I Drug Metabolism

Examples of genetically polymorphic Phase-I enzymes are listed that catalyzed drug metabolism, including selected examples of drugs that have clinically relevant variations in their effects (adopted from Evans & McLeod, 2003).

Drug-Metabolizing Enzyme	Frequency of Variant Poor-Metabolism Phenotype	Representative Drugs Metabolized	Effect of Polymorphism
Cytochrome P-450 2D6 (CYP2D6)	68% in Sweden 1 % in China	Debrisoquin Sparteine Nortriptyline Codeine	Enhanced drug effect Enhanced drug effect Enhanced drug effect Decreased drug effect
Cytochrome P450 2C9 (CYP2C9)	Approximately 3% in England (those homozygous for the *2 and *3 alleles)	Warfarin Phenytoin	Enhanced drug effect Enhanced drug effect
Cytochrome P-450 2C19 (CYP2C19)	2.7% among white Americans 3.3% in Sweden 14.6% in China 18% in Japan	Omeprazole	Enhanced drug effect
Dihydropyrimidine Dehydrogenase	Approximately 1% of Population is heterozygous	Fluorouracil	Enhanced drug effect
Butyrylcholinesterase (pseudocholinesterase)	Approximately 1 in 3500 Europeans	Succinylcholine	Enhanced drug effect

B. Pharmacogenetics of Phase-II Drug Metabolism

Examples of genetically polymorphic Phase-II (conjugating) enzymes are listed that catalyzed drug metabolism, including selected examples of drugs that have clinically relevant variations in their effects. (adopted from Evans & McLeod, 2003).

Drug-Metabolizing Enzyme	Frequency of Variant Poor-Metabolism Phenotype	Representative Drugs Metabolized	Effect of Polymorphism
N-Acetyltransferase 2	52% among white Americans 17% of Japanese	Isoniazid Hydralazine Procainamide	Enhanced drug effect
Uridine diphosphate-glucuronosyltransferase 1A1 (TATA-box polymorphism)	10.9% among whites 4% of Chinese 1% of Japanese	Irinotecan Bilirubin	Enhanced drug effect Gilbert's syndrome
Thiopurine S-methyltransferase	Approximately 1 in 300 whites Approximately 1 in 2500 Asians	Mercaptopurine Azathioprine	Enhanced drug effect. (toxicity)
Catechol O-methyltransferase	Approximately 25% of whites	Levodopa	Enhanced drug effect

Inductions of Phase II by multi-functional inducers including many of the flavanoids molecules found in fruits and vegetables. This

increase in Phase II supports better detoxification and maintain or healthy balance between Phase I and Phase II authorities. At

Table 2
Major Xenobiotic-Metabolizing Enzymes

CYTOCHROMES P450	
1.	Found in bacteria, yeast, plants, animals, insects.
2.	Membrane-bound-usually ER and mitochondrial.
3.	Haemoproteins
4.	Many different isozymes, grouped into families and subfamilies e.g. enzyme-P450 1A1, gene CYP1A1
ALDEHYDE/KETONE REDUTASES	
1.	Found in all special studied.
2.	Reduce aldehyde/ketone to alcohol group.
3.	Two families- carbonyl reductases- aldo- keto reductases.
4.	Carbonyl reductases are related to alcohol dehydrogenises.
5.	Aldo- keto reductases grouped into 7 families AKR1 to AKR7.
GLUTATHIONE S- TRANSFERASES	
1.	Found in animals, plants, and insects.
2.	Conjugates glutathione to electrophilic centres.
3.	Glutathione is gamma- glutamyl- cysteinyl- glycine.
4.	Small proteins made up of two subunits- dimers.
5.	Multiple enzymes grouped into families e.g. GSTA5, GSTP1.
SULFOTRANSFERASES	
1.	Found in mammals.
2.	Conjugates sulfate to compounds.
3.	Uses 3' – phosphoadenosine 5'- phosphosulfate PAPS.
4.	Cytosolic and membrane- bound.
5.	Only different enzymes in human.
UDP-glucuronyl transferase	
1.	Only found in vertebrates
2.	Covertly links sugars (glucuronate) to molecules
3.	Substrate is UDP-glucuronic acid
4.	Membrane-bound (ER and nuclear)
5.	Multiple enzymes e.g. at least 8 in human
6.	Grouped into families e.g. UGT1 group
N-ACETYLTRANSFERASES	
1.	Found in mammals
2.	Located in liver cytosol
3.	Catalyze transfer of acetyl group from acetyl coenzyme A
4.	Only 2 different genes in human
5.	e.g. NAT1 and NAT2.

least in part the enhancement of Phase II activities by fruits and vegetable to protect against many cancers (Cascorbi *et al.*, 1995).

The net result of metabolism, the

metabolic products are often inactive or less active than the parent drug. Some metabolites that contain reactive oxygen species, such as oxygen or hydroxy radicals (Free radicals) are more toxic than parent compound, and may

lead to adverse drug reaction and form DNA and protein adducts, which initiate and promote cancer (Vogel and Motulsky, 1986). Therefore biotransformation either resulting in detoxification or bio-activation of xenobiotics, depends upon the relative activity of the parent drug and its metabolite. Nebert and Dieter (2000) presented a detailed review describing from evaluation of drug metabolic to its present drug understanding after human genome application.

Genetic polymorphism of Drug Metabolizing Enzymes:

The genetic polymorphism could be referred as a monogenic trait, resulting in two non-rare phenotypes (each occurring at a frequency greater than 1%), existing in a normal population (Meyer, 2000).

Lack of effectiveness (no or lower than normal response) or excessive response (even leading to adverse reaction) to drugs given in the usual dosage, has been found as a result of inherited genetic polymorphism in the drug metabolizing enzymes (DMEs). Variation in DNA sequence, due to insertion/deletion or substitution of a single base (SNP) or stretches of bases or repetitive random repeats, micro satellites or Alu segments, result in genetic polymorphism (Stryer, 1981). Variation in amount and or activity of specific DMEs is manifested by inter-individual variation, in drug metabolism. These variations are caused by genetic and environmental factors (Table 1a and 1b). The pharmacogenetic polymorphism, compared to genetic polymorphism could be expressed as a monogenic trait that is caused by the presence, in the same population, of more than one allele, at the same locus; and more than one phenotype in relation to drug interaction with the organism; where as the frequency of the least common allele is greater than 1% (Meyer, 1992).

Distinct populations are found, with individuals differing in their ability of a certain biotransformation reaction, due to genetic polymorphism in DMEs. These individual could be grouped as slow or poor, intermediate

or extensive and rapid or ultra rapid metabolizers as referred to various DME related categories. The slow or poor metabolizer phenotype is monogenically inherited as an autosomal recessive trait, due to absence or marked decrease in the amount and or activity of the enzyme. The intermediate or extensive metabolizers are heterozygous for the alleles, while the wild-type lack the mutation (Meyer and Zanger, 1997).

Genetic polymorphism and the diversity of human genes Genetics:

Extensive genetic variations could occur at several different levels in most populations; including, color, chromosomal structuring and protein characteristics.

Polymorphism is rather common in human. It has been observed that in each individual about 20% of the proteins and so the genes exist in a different form, from the majority of population (Weaver and Hedrick, 1997). In another study out of 71 human genes, 28% were polymorphic and the average heterogenicity was 0.067 (Meyer, 2000). In a population the allelic frequencies may follow the Hardy-Weinberg law, that states: under certain conditions of stability, both allelic frequencies and genotypic ratios remain constant from generation to generation in sexually reproductive populations (Stryer, 1981).

Mutations in the genes that code for the specific biotransformation enzyme causes the polymorphism in DMEs. It follows, generally the autosomal trait, that mutation is not sex linked and that one mutated allele does not express the phenotype, when present with a normal, non mutated (dominant) allele. There are several ways in which genes can be mutated; substitution (where a nucleotide is changed e.g., C to T), insertion or deletion of a base (Hein *et al.*, 2000).

Point mutations refer to changes in one or few bases. Larger changes can also exist, like deletion of entire gene or duplication of the entire gene. Point mutations be silent (without

any consequences at the protein level). While others affect amino acid sequence and thus effecting the biological function of protein. Through many mechanism a point mutations can influence amino acid sequence and or expression. In the missense mutation, a base change alters the sense of a code from one amino acid to another. Some mutation may introduce a stop code. Insertion or deletion may lead to a frame shift, causing incorrect reading of the base sequence (as reading frame has been shifted). A more frequently occurring mutation is incorrect splicing (splicing defect, incorrect removal of introns) (Hein *et al.*, 2000 and Kalow *et al.*, 1986).

Clinical Consequences of Genetic polymorphism of DME:

Motulsky in 1957, for the first time suggested that certain adverse drug reaction could be related to genetically determined variation of drug metabolizing enzyme in the liver (Bertilsson *et al.*, 1993) when he encountered response to succinyl choline (a muscle relaxant). From here the genetic concept in pharmacology begins. At about the same time, in 1960, it was discovered that the metabolism of isoniazid was under genetic control and slow and rapid metabolizers were identified (Price Evans *et al.*, 1960). These laid the principles for Pharmacogenetics effects. Earlier it was referred as isoniazid-acetylase gene, that today we know that, this large inter-individual difference is associated with polymorphism in the NAT2 gene, for which several allelic variants exist that code for non-functional enzymes (Cascorbi *et al.*, 1995). Since then, many clinically important polymorphism have been discovered; including:

Butyl cholinesterase, methyl transferase, UDP-glucurono-syltransferase, sulfotransferase, alcohol dehydrogenase, cytochrome P450 enzymes specially CYP2D6 CYP2C19, CYP2C9 etc., glutathione-s-transferase, in addition to N-acetyltransferase 1 and 2 (Hein *et al.*, 2000; Price Evans *et al.*, 1960 and Cascorbi *et al.*, 1995) (Table 2).

The clinical consequences in case of ultra rapid metabolic phenotype, may result in a therapeutic failure, due to fast elimination of the drug from the body before it can attain therapeutic plasma concentration (Bertilsson *et al.*, 1993). While poor metabolizers may exhibit serious side effects and adverse reactions.

The reverse situation is manifested, in case, if therapeutic effect is inherent in the metabolite, that requires administration of a prodrug, causing therapeutic failure in poor metabolizers and toxicity in rapid phenotypes.

Drug with narrow therapeutic margin, such as chemotherapeutic agents, a slight change in plasma concentration can result in life threatening situation.

It is therefore imperative, that the analysis of genetic polymorphism of DMEs has a great value for the desired therapeutic efficacy, as well as, avoiding toxicity of drugs and/or adverse reaction (Ingelman-Sundberg *et al.*, 1999).

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